



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2010
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #1

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each question is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 60

- Attempt questions 1 – 3
- All questions are of equal value.

Examiner: *A.M.Gainford*

Question 1. (Start a new page.) (20 marks)

- | | Marks |
|--|--------------|
| (a) For the complex number $z = -2 + 2\sqrt{3}i$ find: | 2 |
| (i) $ z $ | |
| (ii) $\arg z$. | |
| (b) Given $w = 5 + 12i$ express each of the following in the form $a + ib$ (for real a and b). | 2 |
| (i) $\bar{w}$ | |
| (ii) w^{-1} | |
| (c) Find the square roots of $32 - 24i$, giving your answer in the form $x + iy$. | 2 |
| (d) Find in modulus-argument form all complex numbers z such that $z^3 + 1 = 0$, and plot them on the Argand diagram. | 3 |
| (e) Sketch (on separate diagrams) the region in the Argand diagram containing the points z for which: | 6 |
| (i) $0 \leq \arg(z - i) \leq \frac{\pi}{6}$ | |
| (ii) $ z - 2i < 2$ | |
| (iii) $\arg\left(\frac{z - i}{z - 1}\right) = \frac{\pi}{2}$ | |
| (f) Solve the equation $z^5 + z = 0$ over the field of complex numbers. | 2 |
| (g) If α, β, γ are the roots of the equation $2x^3 + 5x^2 - 4x + 2 = 0$, find a cubic equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$, and hence or otherwise state the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. | 3 |

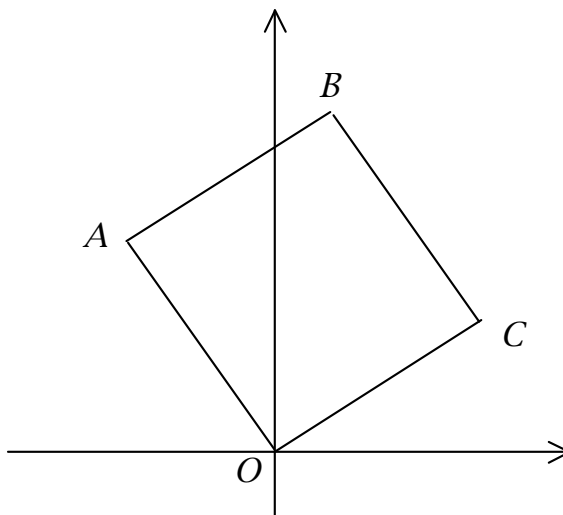
Question 2. (Start a new page.) (20 marks)

Marks
3

- (a) In the Argand diagram at right $OABC$ is a square. The point A represents the complex number $-3 + 4i$.

Find in the form $a + ib$:

- (i) the complex number represented by C
- (ii) the complex number represented by B
- (iii) the complex number represented by the vector CA



- (b) Find $\frac{d}{dx}(x \sin^{-1} x)$, and hence find $\int \sin^{-1} x \, dx$.

3

- (c) Given that $1 + i$ is a root of the equation $x^4 - x^3 - 6x^2 + 14x - 12 = 0$:

4

- (i) explain why $1 - i$ is also a root
- (ii) find all roots of the equation.

- (d) Prove that $\cos^{-1} x + \sin^{-1} x$ is a constant (independent of x).

2

- (e) Solve the quadratic equation $x^2 - (2 + 2i)x - 1 + 2i = 0$.

2

- (e) (i) Determine the domain of the function $y = \sin^{-1}(2x + 1)$. **4**
- (ii) Sketch the graph of $y = \sin^{-1}(2x + 1)$.
- (iii) Solve $\sin^{-1}(2x + 1) = \cos^{-1} x$
- (f) Which complex numbers are the squares of their conjugates? **2**

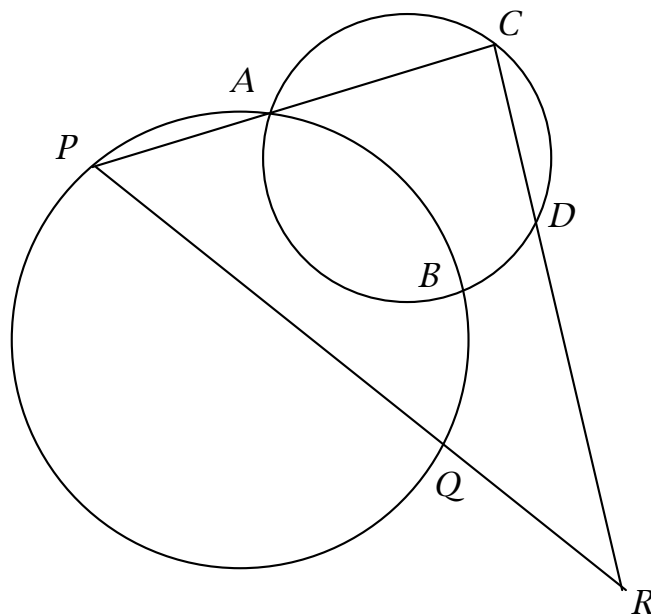
Question 3. (Start a new page.) (20 marks)

- | | | Marks |
|-----|--|--------------|
| (a) | (i) Find the least positive integer k such that $\cos\left(\frac{4\pi}{7}\right) + i \sin\left(\frac{4\pi}{7}\right)$ is a solution of $z^k = 1$. | 2 |
| | (ii) Show that if the complex number w is a solution of $z^n = 1$, then so is w^m , where m and n are arbitrary integers. | 1 |
| (b) | When the polynomial $P(x) = x^3 + 3x^2 + ax + b$ is divided by $x - 1$ the remainder is 5, and when $P(x)$ is divided by $x + 2$ the remainder is 8.

Find the values of a and b . | 2 |
| (c) | (i) Find the co-ordinates of intersection of the curves $y = \cos^{-1} x$ and $y = \sin^{-1} x$. | 4 |
| | (ii) Find the acute angle between the curves $y = \cos^{-1} x$ and $y = \sin^{-1} x$ at the point of intersection. Give your answer in degrees, to the nearest minute. | |
| (d) | (i) Show that if $x = \alpha$ is a double root of the polynomial equation $A(x) = 0$, then $A(\alpha) = A'(\alpha) = 0$. | 1 |
| | (ii) Given that $ax^4 + 4bx + c = 0$ has a double root, prove that $ac^3 = 27b^4$. | 2 |
| (e) | If z is a non-zero complex number such that $z + \frac{1}{z}$ is real, prove that either $\text{Im}(z) = 0$ or $ z = 1$. | 3 |

- (f) In the diagram below PAC , CDR , PQR are straight lines.

5



- (i) Copy the diagram to your answer booklet.
- (ii) Prove that $\angle RQD = \angle RBD$.

This is the end of the paper.

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, n \neq -1; x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, a > 0, -a < x < a$$

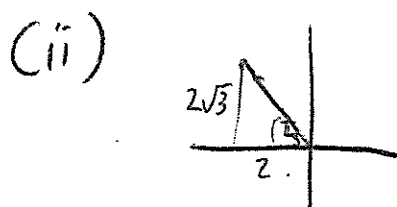
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, x > 0$

QUESTION 1

$$(a) (i) |-2+2\sqrt{3}i| = \sqrt{4+12} = 4.$$



$$\arg(-2+2\sqrt{3}i) = \frac{2\pi}{3}.$$

$$(b) (i) 5-12i$$

$$(ii) \frac{1}{5+12i} \times \frac{5-12i}{5-12i} = \frac{5-12i}{25+144} = \frac{5}{169} - \frac{12}{169}i$$

$$(c) 32-24i = x^2 - y^2 + 2xyi$$

$$32 = x^2 - y^2$$

$$-24 = 2xy \Rightarrow y = -\frac{12}{x} \Rightarrow 144 = x^2 y^2 \Rightarrow y^2 = \frac{144}{x^2}$$

$$32 = x^2 - \frac{144}{x^2}$$

$$x^4 - 32x^2 - 144 = 0$$

$$x^2 = \frac{32 \pm \sqrt{1024 - 4 \times -144}}{2}$$

$$= \frac{32 \pm 40}{2}$$

$$= 36, -4$$

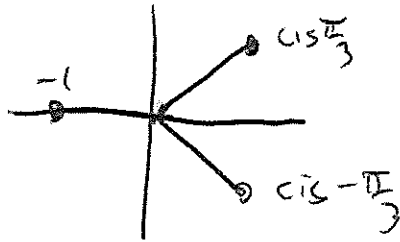
$$x = \pm 6$$

$$\text{when } x=6 \quad y = -2$$

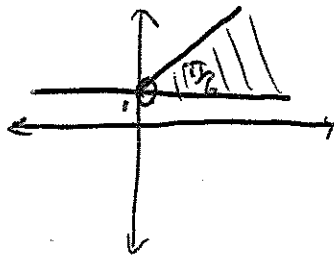
$$\text{or } x=-6 \quad y = 2$$

$$\underline{-6+2i}, \underline{6-2i} \quad 2$$

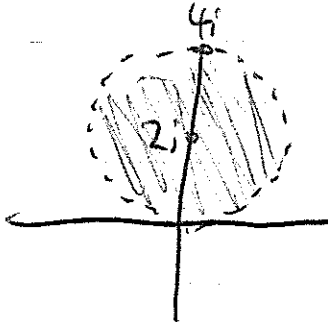
d) $\text{cis } \pm \frac{\pi}{3}, \text{cis } \pi = -1.$



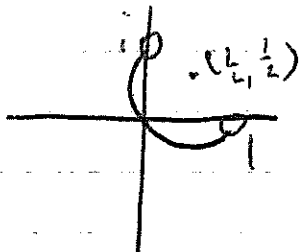
(e) (i)



(ii)



(iii)



(f) $z = 0, \text{cis } \neq \frac{\pi}{4}, \text{cis } \neq \frac{3\pi}{4}.$

(g) $x = \frac{1}{x}$

$$2\left(\frac{1}{x}\right)^3 + 5\left(\frac{1}{x}\right)^2 - 4\left(\frac{1}{x}\right) + 2 = 0.$$

$$2x^3 - 4x^2 + 5x + 2 = 0.$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 2.$$

QUESTION 2

(a) (i) $(-3+4i) \div i = 4+3i$
 (ii) $-3+4i+4+3i=1+7i$
 (iii) $-3+4i-(4+3i)=-7+i$

(b) $\frac{d}{dx}(x \sin^{-1} x) = x \times \frac{1}{\sqrt{1-x^2}} + \sin^{-1} x$
 $= \sin^{-1} x + \frac{x}{\sqrt{1-x^2}}$

$$\int \left(\sin^{-1} x + \frac{x}{\sqrt{1-x^2}} \right) dx = x \sin^{-1} x$$

$$\int (\sin^{-1} x) dx = x \sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}} dx$$

(c) (i) If a complex number is a factor of a polynomial with real coefficients, its conjugate is also a factor.

(ii) $(x-1+i)(x-1-i)$ is a factor.
 (x^2-2x+2) is a factor

$$\begin{array}{r} x^2+x-6 \\ x^2-2x+2 \overline{) x^4-x^3-6x^2+14x-12} \\ \underline{x^3-8x^2+14x} \\ x^3-2x^2+2x \\ \underline{-6x^2+12x-12} \\ -6x^2+12x-12 \\ \underline{} \end{array}$$

Roots $1+i, 1-i, -3, 2$

(d)

$$a = \cos^{-1} x, b = \sin^{-1} x$$

$$\cos a = x, \sin b = x$$

$$\cos a = \sin b$$

$$a+b = \frac{\pi}{2}$$

$$\cos^{-1} x + \sin^{-1} x = \frac{\pi}{2} - \text{constant}$$

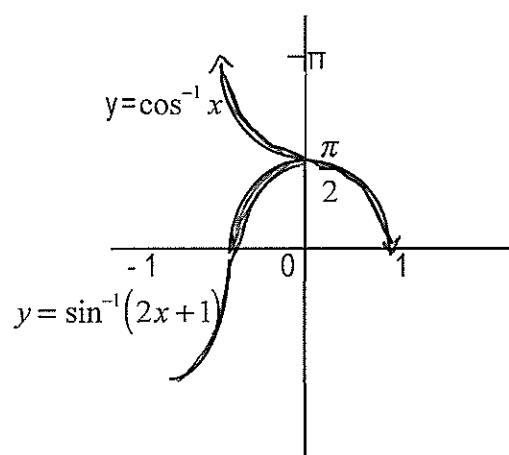
(e) $x = \frac{2+2i \pm \sqrt{(-2-2i)^2 - 4 \times 1 \times (-1-2i)}}{2}$
 $= \frac{2+2i \pm \sqrt{4+8i-4+4-8i}}{2}$

$$= \frac{2+2i \pm 2}{2}$$

$$x = 2+i \text{ or } i$$

(f) (i)

$$\begin{aligned} \text{Domain } -1 \leq 2x+1 \leq 1 \\ -2 \leq 2x \leq 0 \\ -1 \leq x \leq 0 \end{aligned}$$



(f) (ii) From graph $x=0$

(g)

$$x+iy = (x-iy)^2$$

$$= x^2 - 2xyi - y^2$$

$$(1) x = x^2 - y^2$$

$$y = -2xy$$

$$1 = -2x$$

$$x = -\frac{1}{2}$$

$$\text{From (1)} -\frac{1}{2} = \frac{1}{4} - y^2$$

$$y = \pm \frac{\sqrt{3}}{2}$$

$$\text{nos. are: } 0, 1, -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

Question (3)

(a) (i) $z^k = 1$

Let $z = cis \theta$

$z^k = cis k\theta + i \sin k\theta$

Equate real part

$cis k\theta = 1$

$\Rightarrow k\theta = 2n\pi$ [2]

$\therefore \theta = \frac{2n\pi}{k}$

i.e. $\theta = 0, \frac{2\pi}{k}, \frac{4\pi}{k}, \dots$

$\Rightarrow k = 7$

i.e. the least positive integer.

(ii) w is a solution to $z^4 = 1$

$\therefore w^4 = 1$

Now $w^m = (w^4)^m$

$= 1^m$

$= 1$ [1]

$\Rightarrow w^m$ is also a solution

(b) $P(x) = x^3 + 3x^2 + ax + b$

$P(1) = 5$

$P(-2) = 8$

$5 = 1 + 3 + a + b$ [2]

$8 = -8 + 12 - 2a + b$

$\therefore a + b = 1$ — (1)

$-2a + b = 4$ — (2)

(1) - (2) $3a = -3$

$\therefore a = -1$ — (3)

Subst (3) into (1)

$\therefore b = 2$ — (4)

(c) (i) $\cos^{-1} x = \sin^{-1} x$

i.e. $\sin^{-1} x - \cos^{-1} x = 0$ [1]

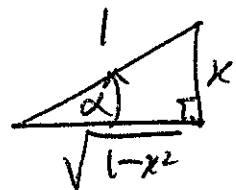
Let $\alpha = \sin^{-1} x, \frac{-\pi}{2} \leq \alpha \leq \frac{\pi}{2}$
 $\beta = \cos^{-1} x, 0 \leq \beta \leq \pi$

$\Rightarrow \sin \alpha = x$

$\cos \alpha = \sqrt{1-x^2}$

$\cos \beta = x$

$\sin \beta = \sqrt{1-x^2}$



$\alpha - \beta = 0$

Take $\sin(\alpha - \beta) = 0$ [2]

$\sin \alpha \cos \beta - \cos \alpha \sin \beta = 0$

$x^2 - (\sqrt{1-x^2})^2 = 0$

$2x^2 = 1, x^2 = \frac{1}{2}$

$x = \pm \frac{1}{\sqrt{2}}, \therefore x = \frac{1}{\sqrt{2}}$

$\therefore (\frac{1}{\sqrt{2}}, \pi/4) \quad y = \pi/4$

(c)

$$(ii) \quad y_1 = \cos^{-1} x \quad \boxed{4}$$

$$m_1 = \frac{dy_1}{dx} = -\frac{1}{\sqrt{1-x^2}}$$

$$m_2 = \frac{dy_2}{dx} = \frac{1}{\sqrt{1-x^2}}$$

$$\left. \frac{dy_1}{dx} \right|_{x=\frac{1}{\sqrt{2}}} = \frac{-1}{\frac{1}{\sqrt{2}}} = -\sqrt{2}$$

$$\left. \frac{dy_2}{dx} \right|_{x=\frac{1}{\sqrt{2}}} = \frac{1}{\frac{1}{\sqrt{2}}} = \sqrt{2}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|^2$$

$$= \left| \frac{-\sqrt{2} - \sqrt{2}}{1 + (-2)} \right|$$

$$= 2\sqrt{2}$$

$$\therefore \theta = \tan^{-1} 2\sqrt{2} = 70.32^\circ$$

(d) If $x = \alpha$

(i) is a double root of $A(x)$

then

$$A(x) = (x - \alpha)^2 Q(x)$$

$$\therefore A(\alpha) = (\alpha - \alpha)^2 Q(\alpha) = 0$$

$$A'(x) = 2Q(x)(x - \alpha) + (x - \alpha)^2 Q'(x)$$

$$\therefore A'(\alpha) = 2Q(\alpha)(\alpha - \alpha) + (\alpha - \alpha)^2 Q'(\alpha) = 0$$

$$\therefore A(\alpha) = A'(\alpha) = 0 \quad \boxed{1}$$

$$(ii) \quad p(x) = ax^4 + 4bx + c$$

$$p(\alpha) = 0 \text{ and } p'(\alpha) = 0$$

$$a\alpha^4 + 4b\alpha + c = 0 \quad \text{--- (1)}$$

$$\text{and } 4a\alpha^3 + 4b = 0 \quad \text{--- (2)}$$

$$\text{From (2)} \quad \alpha^3 = -\frac{b}{a}$$

$$\text{--- (3)}$$

From (1)

$$a(\alpha^3)(\alpha) + 4b(\alpha) + c = 0$$

$$\text{--- } b\alpha + 4\alpha = -c$$

$$\therefore \alpha = -\frac{c}{3b}$$

$$\text{--- (4)}$$

Subst (4) into (1)

$$a\left(\frac{c^4}{81b^4}\right) - 4b\left(\frac{c}{3b}\right) + c = 0$$

$$\Rightarrow ac^3 = 27b^4 \quad \boxed{2}$$

$$(e) \quad \text{Let } z = x + iy$$

$$\frac{1}{z} = \frac{x - iy}{x^2 + y^2} \quad \boxed{3}$$

$$\therefore z + \frac{1}{z} = x + iy + \frac{x - iy}{x^2 + y^2}$$

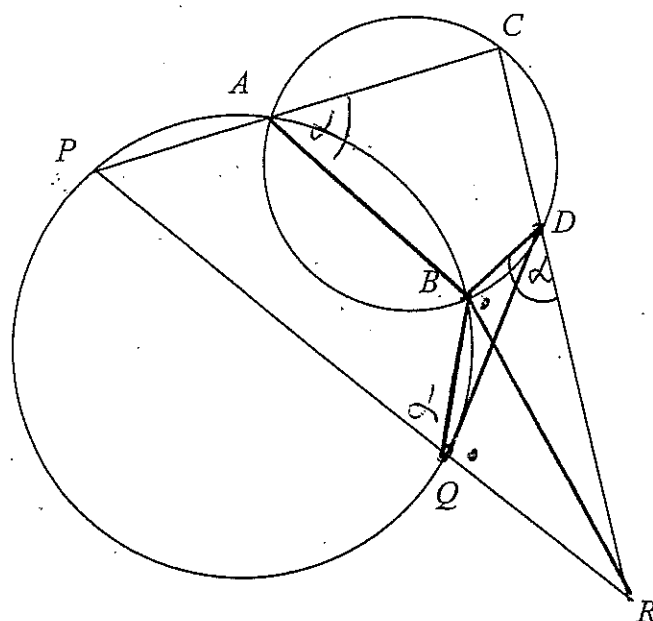
If $z + \frac{1}{z}$ is real

$$\Rightarrow \text{Im}\left(z + \frac{1}{z}\right) = 0$$

$$iy\left(1 - \frac{1}{x^2 + y^2}\right) = 0$$

$$\Rightarrow y = 0 \text{ or } x^2 + y^2 = 1$$

$$\text{i.e. } |z|^2 = 1$$



5

(i) Copy the diagram to your answer booklet.

(ii) Prove that $\angle RQD = \angle RBD$.

Let $\angle PQB = \alpha \therefore \angle CAB = \alpha$.

(Exterior angle of cyclic quad ABQP,

Similarly, $\angle BDR = \alpha$.

3 In quad. BDRQ

$\angle BDR = \angle PQB = \alpha$.

$\therefore$ BDRQ is a cyclic quad

$\therefore \angle RQD = \angle RBD$

2 (Converse of angles in the same segment).